Lesson Planning

Complete Lesson Plan Book

Class: Class 9

Subject: Physics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Measurements

Learning Objectives

• Define a physical quantity and distinguish between base and derived quantities.

• State the seven SI base units and use common prefixes correctly.

• Explain the need for significant figures in measured values.

• Use measuring instruments such as the vernier caliper and screw gauge.

Key Concepts

• Physical quantity: a quantity that can be measured

• Base quantities (length, mass, time) vs derived quantities (speed, area)

• SI base units: metre, kilogram, second, ampere, kelvin, mole, candela

• Prefixes: kilo, mega, milli, micro and their powers of ten

• Significant figures and rules for counting them

Teaching Methodology (Step by Step)

1. Ask students how they would measure the thickness of one page of a book to create curiosity.
2. Introduce base and derived quantities with daily-life examples on the board.
3. Demonstrate the vernier caliper and screw gauge; let volunteers read a measurement.
4. Give short numerical practice on unit conversion and significant figures, then check answers.

Classroom Activities

• Measure classroom objects (desk length, notebook thickness) with a ruler and record readings.

• Unit-conversion relay: teams convert values like 2.5 km to mm on the board.

Assessment Questions

1. What is the difference between a base and a derived quantity?
2. Name the seven SI base quantities with their units.
3. Convert 3.6 km/h into m/s, showing all steps.
4. How many significant figures are there in 0.00450?

Homework

• Complete the worksheet on SI prefixes and significant figures.

• Measure five objects at home and write each reading with correct significant figures.

Time Allocation

5 periods of 40 minutes each

Unit 2: Kinematics

Learning Objectives

• Define rest, motion, distance, displacement, speed and velocity.

• Distinguish between uniform and non-uniform motion.

• Define acceleration and solve numerical problems using equations of motion.

• Plot and interpret distance-time and velocity-time graphs.

Key Concepts

• Rest vs motion; scalar vs vector quantities

• Distance (scalar) vs displacement (vector)

• Speed, velocity and their units

• Acceleration: positive, negative and zero

• Equations of motion: v = u + at, S = ut + 1/2 at^2, 2aS = v^2 - u^2

Teaching Methodology (Step by Step)

1. Roll a ball across the desk and ask: when is it at rest and when in motion?
2. Explain distance vs displacement by walking two different paths to the same point.
3. Derive the three equations of motion step by step on the board.
4. Solve two numericals together, then let students solve one independently.

Classroom Activities

• Draw distance-time graphs for a walk to school and interpret them in pairs.

• Time a toy car over a fixed distance and calculate its average speed.

Assessment Questions

1. Define velocity. How is it different from speed?
2. A car starts from rest and reaches 20 m/s in 5 s. Find its acceleration.
3. Sketch a velocity-time graph for uniform acceleration and label its slope.

Homework

• Solve five textbook-style numericals on equations of motion.

• Draw and label graphs for uniform velocity and uniform acceleration.

Time Allocation

6 periods of 40 minutes each

Unit 3: Dynamics

Learning Objectives

• State the three laws of motion with one example of each.

• Define momentum and state the law of conservation of momentum.

• Define friction and explain methods to reduce or increase it.

• Solve numerical problems using F = ma.

Key Concepts

• First law of motion: inertia

• Second law of motion: F = ma

• Third law of motion: action and reaction pairs

• Momentum: p = mv; conservation of momentum

• Friction: advantages, disadvantages and control

Teaching Methodology (Step by Step)

1. Push a book on the desk and ask why it stops: introduce friction and inertia.

2. Demonstrate action-reaction with two students on chairs pushing each other.

3. Explain momentum with the example of a truck vs a bicycle at the same speed.

4. Solve numericals on F = ma and momentum, then give a short class quiz.

Classroom Activities

• List five examples of the third law of motion seen at home.

• Roll balls of different masses down a ramp and compare their motion.

Assessment Questions

1. State the second law of motion with its mathematical form.

2. Why do we wear seat belts? Answer using the first law.

3. A 2 kg mass accelerates at 3 m/s^2. Find the applied force.

4. Give two methods to reduce friction and two to increase it.

Homework

• Write one daily-life example of each law of motion with explanation.

• Solve the numerical exercise on momentum from the notebook.

Time Allocation

6 periods of 40 minutes each

Unit 4: Turning Effect of Forces

Learning Objectives

• Define moment of force (torque) and state its unit.

• Explain the principle of moments with examples.

• Define centre of mass and centre of gravity.

• Describe stable, unstable and neutral equilibrium.

Key Concepts

• Moment of force: force x perpendicular distance

• Clockwise and anticlockwise moments

• Principle of moments: sum of clockwise = sum of anticlockwise

• Centre of mass vs centre of gravity

• States of equilibrium and their conditions

Teaching Methodology (Step by Step)

1. Open a door near the hinge and far from it: ask which is easier and why.
2. Balance a metre rod on a finger to locate its centre of gravity.
3. Demonstrate the principle of moments with a simple see-saw model.
4. Show stable vs unstable equilibrium using a cone in different positions.

Classroom Activities

• Find the centre of gravity of an irregular cardboard shape by hanging it.

• Balance a ruler on a pencil and add coins to test the principle of moments.

Assessment Questions

1. Define torque and write its unit.
2. State the principle of moments.
3. Why is the base of a racing car kept low and wide?
4. Distinguish between centre of mass and centre of gravity.

Homework

• Draw labelled diagrams of the three states of equilibrium.

• Solve numericals on moments of force.

Time Allocation

5 periods of 40 minutes each

Unit 5: Gravitation

Learning Objectives

• State the law of gravitation with its mathematical form.

• Explain the variation of 'g' with altitude and depth.

• Define mass and weight and distinguish between them.

• Explain artificial satellites and their uses.

Key Concepts

• Law of gravitation: F = G m1 m2 / r^2

• Gravitational constant G and its value

• Acceleration due to gravity 'g' and its variation

• Mass (kg) vs weight (N): W = mg

• Satellites: geostationary and polar orbits

Teaching Methodology (Step by Step)

1. Drop a stone and a crumpled paper together: discuss what pulls them down.

2. Explain the inverse-square law with a simple diagram on the board.

3. Compare mass and weight using a spring balance and a beam balance.

4. Discuss satellite uses: weather, communication, GPS.

Classroom Activities

• Calculate your weight on the Moon (g = 1.6 m/s^2) and compare with Earth.

• Research in pairs: name three Pakistani satellites or space efforts.

Assessment Questions

1. State the law of gravitation.

2. Why does 'g' decrease as we go up a mountain?

3. Differentiate between mass and weight with units.

4. What is a geostationary satellite? Give one use.

Homework

• Solve numericals on the law of gravitation.

• Write a paragraph on the uses of artificial satellites.

Time Allocation

5 periods of 40 minutes each

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Unit 6: Work and Energy

Learning Objectives

• Define work and state its unit (joule).

• Define kinetic and potential energy with examples.

• State the law of conservation of energy.

• Define power and solve numericals on work, energy and power.

Key Concepts

• Work: W = Fd (only when force has displacement component)

• Kinetic energy: Ek = 1/2 mv^2

• Potential energy: Ep = mgh

• Interconversion of energy forms

• Power: P = W/t; unit watt

Teaching Methodology (Step by Step)

1. Lift a book to different heights: ask where it has more energy and why.
2. Derive kinetic and potential energy formulas with simple steps.
3. Show energy interconversion with a swinging pendulum.
4. Solve numericals on work, energy and power as guided practice.

Classroom Activities

• Identify energy conversions in five home appliances and list them.

• Measure time to climb stairs and estimate your power output.

Assessment Questions

1. Define work. When is work done said to be zero?
2. Write the formulas of kinetic and potential energy.
3. A 5 kg object falls from 10 m. Find its potential energy (g = 10 m/s^2).
4. State the law of conservation of energy with one example.

Homework

• Solve the chapter-end numericals on energy and power.

• Draw an energy-flow diagram for a hydroelectric power station.

Time Allocation

5 periods of 40 minutes each

Unit 7: Properties of Matter

Learning Objectives

• Define density and solve numericals using density = mass/volume.

• State Archimedes' principle and the law of floatation.

• Explain pressure in fluids with examples.

• Describe elasticity and Hooke's law.

Key Concepts

• Density and its unit kg/m^3

• Pressure: P = F/A; pressure in liquids

• Archimedes' principle: upthrust equals weight of displaced fluid

• Law of floatation

• Elasticity, stress, strain and Hooke's law

Teaching Methodology (Step by Step)

1. Drop an iron nail and a wooden block in water: ask why one sinks.
2. Demonstrate upthrust by weighing an object in air and in water.
3. Explain hydraulic machines using the car-lift example.
4. Stretch a spring to show Hooke's law and plot force vs extension.

Classroom Activities

• Test which household objects float or sink and record the pattern.

• Measure the density of a stone using a measuring cylinder and balance.

Assessment Questions

1. State Archimedes' principle.
2. Why does a ship float while an iron nail sinks?
3. Define pressure and write its unit.
4. State Hooke's law.

Homework

• Solve numericals on density and pressure.

• Write two uses of hydraulic machines in daily life.

Time Allocation

5 periods of 40 minutes each

Unit 8: Thermal Properties of Matter

Learning Objectives

• Define temperature and describe the construction of a thermometer.

• Explain thermal expansion in solids, liquids and gases.

• Describe the three modes of heat transfer.

• Define specific heat capacity and solve simple numericals.

Key Concepts

• Temperature vs heat

• Thermometric scales: Celsius, Fahrenheit, Kelvin

• Linear, superficial and volumetric expansion

• Conduction, convection and radiation

• Specific heat capacity and its unit J/(kg K)

Teaching Methodology (Step by Step)

1. Touch a metal and a wooden desk: ask which feels colder and why.
2. Show expansion with a heated metal rod or the ball-and-ring demo.
3. Demonstrate convection using warm coloured water in a tank.
4. Explain specific heat with the sea-breeze example.

Classroom Activities

• List three examples each of conduction, convection and radiation at home.

• Observe and record how a bimetallic strip bends when heated.

Assessment Questions

1. Name the three modes of heat transfer with one example each.
2. Convert 37 C to Fahrenheit and Kelvin.
3. Why are gaps left between railway tracks?
4. Define specific heat capacity.

Homework

• Solve numericals on temperature conversion and heat capacity.

• Write a short note on the sea breeze and land breeze.

Time Allocation

5 periods of 40 minutes each